
Time-Aware Transformers for Longitudinal MRI in Alzheimer’s Disease Progression Prediction

Code: <https://github.com/ahmad-sarfraz345/mci2ad-survival-transformer>

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http://adni.loni.usc.edu/wp-content/uploads/how_to_apply/ADNI_Acknowledgement_List.pdf

Abstract

Predicting the conversion from mild cognitive impairment (MCI) to Alzheimer’s disease (AD) is a clinically consequential task, yet existing deep learning methods often treat it as a static classification problem. Current longitudinal models struggle with the temporally irregular nature of real-world clinical MRI sequences, frequently relying on assumed fixed intervals or heuristic temporal decay mechanisms (Hong et al., 2020). Using longitudinal MRI data from the ADNI (Alzheimer’s Disease Neuroimaging Initiative, 2025) cohort, we propose a Time-Aware Transformer architecture that directly models structural brain change trajectories from irregular multi-visit MRI data. Our approach introduces two key mechanisms to model irregular temporal sampling: (1) explicit continuous temporal encoding to model elapsed time between visits, and (2) Attention with Linear Biases (ALiBi) to penalize attention scores based on elapsed time between scans. Evaluated on the ADNI (Alzheimer’s Disease Neuroimaging Initiative, 2025) dataset, our standard 3D CNN-Transformer baseline achieved a Concordance Index (C-index) of 0.752. Incorporating temporal encoding improved the C-index to 0.766, and the final integration of ALiBi established our best performance with a C-index of 0.777. Furthermore, our time-aware architecture significantly enhanced short-term prognostic accuracy, achieving an AUC@2y of 0.958 compared to the baseline’s

0.919. These results demonstrate that explicitly encoding temporal gaps improves the ability of Transformers to leverage irregular longitudinal scans for AD progression prediction.

1. Introduction

Alzheimer’s disease is characterized by a gradual deterioration of cognitive function before symptoms become apparent. MCI (Mild Cognitive Impairment) is the stage between a normal brain and dementia where some patients remain stable for years while others rapidly progress to dementia. Early identification of likely converters and prediction of their probable conversion pattern is one of the most consequential tasks in computational neuroscience.

Most of the work done in deep learning-based AD research has framed it as a static image classification problem, using single MRIs. However, several recent works attempted to frame it as a progression prediction with notable results. Aghaei and Ebrahimi Moghaddam (Aghaei & Ebrahimi Moghaddam, 2025) chained a ViT-GAN that synthesized a two-year follow-up scan with a 3D-CNN, and achieved strong accuracy on CN (cognitively normal) to AD prediction. However, fixing the temporal window to a single synthetic interval prevents accurate trajectory modelling. Shetty et al. (Shetty et al., 2025) achieved a 95% accuracy on pMCI vs sMCI classification by feeding SPM-localized voxel patches into a TimeDistributed 3D-CNN. However the model sees only one baseline MRI and longitudinal imaging plays no role. Chattopadhyay et al. (Chattopadhyay et al., 2026) expands it by framing it as a regression problem across 2319 participants from three cohorts, predicting continuous cognitive decline scores. However the temporal signal is encoded only in the outcome variable instead of the imaging input. Sun et al. (Sun et al., 2026) are among the few works that explicitly process multi-timepoint structural MRI (scans at baseline, 12 months and 24 months) yet leaves out MCI patients, focusing only on CN vs. AD, even though MCI is where trajectory information is most actionable.

A parallel line of work has incorporated temporal modelling through recurrent architectures. Nguyen et al. (Nguyen et al., 2020) and Cui et al. (Cui et al., 2019) applied LSTM and GRU networks to longitudinal multimodal data and demonstrated that sequence models can extract useful temporal signals over classical baselines. Aghajanian et al. (Aghajanian et al., 2025) advanced this with a time-aware LSTM that incorporated elapsed time between visits through a scalar decay function, reporting a C-index of 0.91. However, this approach uses only 3 visits and the authors acknowledge that the scalar decay cannot capture nonlinear temporal dynamics. Nawaz et al. (Nawaz et al., 2026) similarly achieved high classification accuracy with a bidirectional GRU. However they treat all timepoints as uniformly spaced regardless of time elapsed. Across all of these, temporal modelling was restricted to short fixed length sequences with either regular sampling assumed or irregularity handled through approximation.

This intersection remains unoccupied is a longitudinal deep learning architecture applied specifically to MCI patients, operating on irregularly sampled scan sequences that explicitly models brain change trajectories to predict conversion. This gap is consistently acknowledged in the future work sections of all the contributions above.

Our paper addresses this gap. Our contributions are as follows:

1. We established a longitudinal baseline pairing a 3D CNN with a Transformer sequence model, trained under Cox partial likelihood on ADNI (Alzheimer’s Disease Neuroimaging Initiative, 2025), evaluated with the C-index and time-windowed AUC as primary metrics.
2. We replaced visit index positional encoding with continuous temporal encoding that represents both elapsed time and intervisit gap in years. This enables the model to distinguish between clinically distinct progression windows.
3. We introduced an ALiBi-inspired temporal attention bias that directly penalizes attention scores by real elapsed time between visit pairs, with learnable per-head slopes. This allowed different attention heads to specialize to different temporal scales.

Each modification was applied iteratively, and together they produced consistent improvements across the primary survival metrics. These modifications improved the C-index from 0.752 to 0.766 and finally to 0.777, alongside improvements in short-horizon AUC.

2. Methodology

2.1. Problem Formulation

We formulate MCI-to-AD conversion prediction as a longitudinal survival analysis problem using irregularly sampled structural MRI sequences. For each patient i , the input consists of a sequence of T_i preprocessed 3D MRI volumes:

$$\{x_{i,1}, x_{i,2}, \dots, x_{i,T_i}\}$$

acquired at timestamps:

$$\{t_{i,1}, t_{i,2}, \dots, t_{i,T_i}\}$$

where the intervals between visits are irregular. Although the original dataset contains variable-length trajectories, we standardized all sequences to the first four visits for consistency across subjects.

The objective is to estimate a scalar log-risk score representing the likelihood of progression from mild cognitive impairment (MCI) to Alzheimer’s disease (AD). Rather than treating conversion as a binary classification problem, we model it within a survival analysis framework using the Cox proportional hazards objective. The Cox objective is particularly suitable because it models relative risk ordering while naturally handling censored observations. This formulation allows the model to account for censored patients and differing follow-up durations.

Converters ($n = 155$) are assigned ground-truth conversion times obtained from the ADNI (Alzheimer’s Disease Neuroimaging Initiative, 2025) clinical records, while stable MCI subjects ($n = 256$) are treated as right-censored observations. The dataset is divided using a patient-level stratified split to preserve class balance across subsets, resulting in 287 training subjects, 61 validation subjects, and 63 test subjects, with a converter ratio of approximately 38% in all splits.

We report time-dependent AUC at horizons $t \in \{2, 3, 5\}$ years using the cumulative/dynamic ROC formulation computed from test-set risk scores and survival outcomes. Censoring is handled via inverse-probability weighting using the training-set survival distribution, matching the `cumulative_dynamic_auc` estimator (730/1095/1825 days). These AUC@t values summarize discriminative ability at each horizon while properly accounting for right-censored subjects.

2.2. Dataset and Preprocessing

Data used in the preparation of this article were obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (<http://adni.loni.usc.edu>). The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W.

Weiner, MD. The original goal of ADNI was to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early Alzheimer’s disease (AD). The current goals include validating biomarkers for clinical trials, improving the generalizability of ADNI data by increasing diversity in the participant cohort, and to provide data concerning the diagnosis and progression of Alzheimer’s disease to the scientific community.

Experiments were conducted on longitudinal structural MRI data obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) (Alzheimer’s Disease Neuroimaging Initiative, 2025) cohort. The dataset consists of 411 subjects diagnosed with mild cognitive impairment at baseline, including 155 subjects who later converted to Alzheimer’s disease and 256 subjects who remained stable throughout the observation period.

MRI scans were loaded directly from DICOM files and preprocessed into standardized 3D volumes. Each scan was resized and centre-cropped to a spatial resolution of $96 \times 96 \times 96$, followed by per-volume intensity normalization. To ensure sufficient temporal information for longitudinal modelling, only subjects with at least four visits were included. The first four visits of each subject were retained.

The resulting dataset captures irregular visit intervals under a standardized four-visit longitudinal protocol, reflecting realistic clinical follow-up conditions.

2.3. Baseline Architecture

Our baseline model is adapted from (Hu et al., 2023) and it combines a 3D convolutional neural network (CNN) encoder with a Transformer-based temporal aggregation module and a Cox survival prediction head. The architecture is designed to model longitudinal structural brain changes across multiple MRI visits while remaining agnostic to absolute scan timing.

3D CNN Encoder: Each MRI volume is independently processed using a four-block 3D CNN encoder. The encoder progressively extracts hierarchical spatial representations using convolutional layers with channel dimensions $32 \rightarrow 64 \rightarrow 128 \rightarrow 256$. Each block consists of 3D convolution, batch normalization, ReLU activation, and max pooling.

Following the final convolutional stage, adaptive average pooling reduces the spatial representation to a fixed-dimensional feature vector, which is projected to a latent embedding dimension of $d_{\text{model}} = 128$.

Transformer Temporal Encoder: The sequence of visit embeddings is passed to a Transformer encoder consisting

of two layers with four attention heads and feed-forward dimension 256. A learnable CLS token is prepended to the sequence and used as the global patient representation.

The baseline model applies sinusoidal positional encoding based solely on visit index order. Consequently, the model can distinguish between early and late visits in a sequence but has no access to the true elapsed time between scans.

Cox Survival Head: The final CLS embedding is passed through a linear projection layer to produce a scalar log-risk prediction. Training is performed using the Cox partial likelihood loss, enabling survival modelling with censored observations.

The baseline architecture contains approximately 1,461,633 trainable parameters. The complete model pipeline and the progression of improvements are illustrated in Figure 1.

2.4. Improvement 1: Continuous Temporal Encoding

The first improvement introduces explicit modelling of irregular visit timing through a continuous temporal encoding mechanism. In the baseline architecture, positional encoding depends only on visit index and therefore assumes uniform temporal spacing between scans. However, in clinical practice, MRI visits often occur at highly irregular intervals, and these temporal gaps may themselves contain prognostic information.

To address this limitation, we replace the baseline sinusoidal positional encoding with a `ContinuousTimeEncoding` module that operates directly on real elapsed time values. For each visit, two temporal representations are computed: (1) absolute elapsed time from the first visit, and (2) relative time gap from the previous visit. Both representations are encoded using sinusoidal functions and concatenated before being projected back to the Transformer embedding dimension. This allows the model to distinguish between patients with identical visit counts but different temporal progression patterns.

For example, a patient observed over six months and another observed over three years may exhibit similar structural changes but represent fundamentally different disease trajectories. Explicit temporal encoding enables the model to incorporate this distinction directly into the sequence representation.

The addition of the projection layer increases the total trainable parameter count to 1,478,273.

2.5. Improvement 2: ALiBi Temporal Attention Bias

Our second improvement introduces explicit temporal biasing within the Transformer attention mechanism. While the

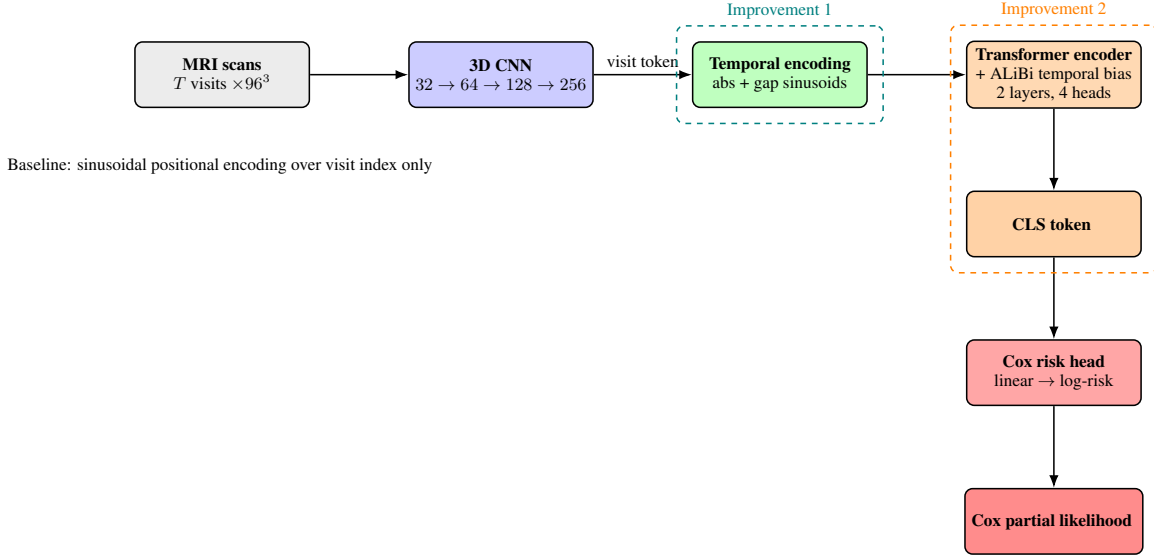


Figure 1. Overall model pipeline. Dashed outlines indicate components introduced in each improvement stage. Improvement 2 stacks on top of Improvement 1

continuous temporal encoding module provides the model with timing information at the embedding level, the self-attention operation itself remains temporally unconstrained.

For a pair of visits (i, j) , the attention bias for attention head h is defined as:

$$-\alpha_h |t_i - t_j|$$

where t_i and t_j denote the visit timestamps and α_h is a learnable slope parameter for the attention head h . The resulting bias is added directly to the raw attention logits before softmax normalization. Larger temporal gaps therefore reduce attention scores proportionally. Figure 2 illustrates an example temporal bias matrix for a sequence of irregularly spaced visits.

The learnable slopes are initialized log-uniformly between 0.5 and 4.0, enabling different attention heads to specialize at different temporal scales. Some heads may prioritize recent visits, while others capture longer-term disease progression patterns.

Importantly, the ContinuousTimeEncoding module from Improvement 1 is retained, making the temporal modelling strategy cumulative rather than replacement-based.

2.6. Hyperparameters and Training Configuration

Table 1 summarizes the primary hyperparameters used across all experiments.

Table 1. Training and architecture hyperparameters

Hyperparameter	Value
Embedding dimension (d_{model})	128
Attention heads	4
Transformer layers	2
Feed-forward dimension	256
Dropout	0.1
Batch size	4
Epochs	30
Learning rate	1×10^{-4}
Weight decay	1×10^{-4}
Gradient clipping	1.0
Early stopping patience	3
LR schedule	Warmup + cosine decay
Optimizer	AdamW

	V1 0y	V2 0.5y	V3 1.5y	V4 3.0y
V1	0.0	-0.5	-1.5	-3.0
V2	-0.5	0.0	-1.0	-2.5
V3	-1.5	-1.0	0.0	-1.5
V4	-3.0	-2.5	-1.5	0.0

Figure 2. Example ALiBi temporal attention bias matrix for one attention head with slope $m_h = 1.0$.

3. Results

Table 2 summarises test-set performance across the three model variants. Training dynamics across all models are shown in Figure 3, while Figure 4 compares the primary survival metrics across model variants. C-index improves monotonically: +0.015 from the Baseline to Improvement 1, and +0.011 from Improvement 1 to Improvement 2. AUC@2y sees the largest jump at Improvement 2 (+0.032), suggesting near-term conversion prediction benefits the most from temporal attention biasing. AUC@5y is non-monotone: it dips at Improvement 1 but recovers at Improvement 2. Binary classification metrics are flat across all models, which suggests that the proposed temporal mechanisms improve the model’s ability to rank patient risk trajectories and estimate conversion timing, even when binary decision boundaries remain unchanged. In particular, the substantial gain in AUC@2y indicates that temporally-aware attention is especially beneficial for short-horizon clinical prognosis.

Table 2. Test-set performance across model variants. $\uparrow$ denotes higher is better.

Metric	Baseline	+ Temporal Enc.	+ ALiBi
C-index $\uparrow$	0.7520	0.7665	0.7770
AUC@2y $\uparrow$	0.9193	0.9259	0.9582
AUC@3y $\uparrow$	0.6926	0.7286	0.7457
AUC@5y $\uparrow$	0.7496	0.7318	0.7405
Binary AUC $\uparrow$	0.7372	0.7404	0.7532
Accuracy	0.6825	0.6825	0.6825
F1	0.6429	0.6429	0.6429
Sensitivity	0.7500	0.7500	0.7500
Specificity	0.6410	0.6410	0.6410

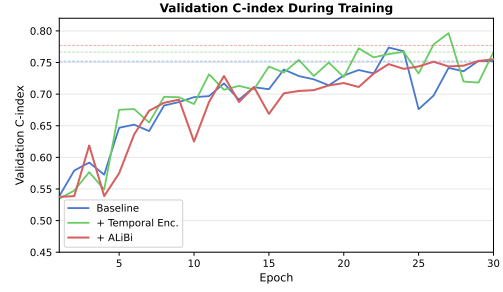


Figure 3. Validation C-index over 30 training epochs for the Baseline, + Temporal Encoding, and + ALiBi models. Dashed horizontal lines mark each model’s best test-set C-index.

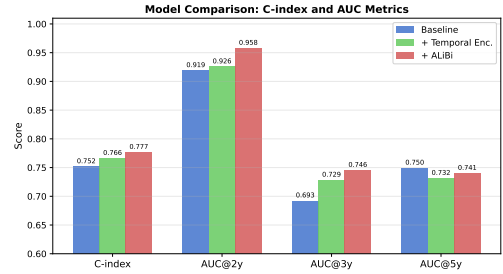


Figure 4. Comparison of C-index and time-dependent AUC at 2, 3, and 5 years across the three model variants.

4. Discussion

4.1. Why Temporal Encoding Helped

The continuous time encoding provides the model with a signal that visit index alone could not convey. For example, a 2-year gap followed by a 6-month scan is a different trajectory than evenly spaced 1-year visits. As shown in Table 2, the AUC@3y improvement (+0.036) showed that medium-term conversion prediction benefited most.

4.2. Why ALiBi Further Helps

The attention bias directly controls which visits attend to which based on real elapsed time. Visits that are temporally close should naturally have stronger structural correlation. ALiBi encodes this as an inductive bias rather than requiring the model to learn it from the data on its own. As shown in Table 2, the AUC@2y jump (+0.032) reflects that short-horizon predictions rely heavily on the most recent visits, and this is emphasised by ALiBi’s recency bias.

4.3. Why Binary Classification Metrics Fall Flat

The proposed improvements optimise risk ranking, not threshold-based classification. The threshold-dependent metrics e.g., accuracy and F1 score, will remain flat unless the model’s calibration at that specific cut-point changes.

This is expected behaviour.

4.4. AUC@5y Non-Monotonic Behaviour

As shown in Table 2, the AUC@5y dip at Improvement 1 is notable. One hypothesis is that temporal encoding may introduce noise at long horizons, where the absolute elapsed time differences between patients are large and variable. However, this should be experimented with and investigated in future work.

4.5. Ablation Note

We tested our improvements iteratively by stacking one on the other. Future work should isolate the contribution of each temporal component through controlled ablation experiments, including ALiBi without continuous temporal encoding and vice versa.

4.6. Compute Constraints and Training Duration

All primary experiments in this work were conducted using a fixed training budget of 30 epochs to ensure fair comparison across model variants under limited computational resources. While this allowed consistent evaluation of the proposed architectural modifications, the models had likely not fully converged at this training duration.

To investigate this, we conducted an additional extended training experiment on the final ALiBi-based model using 50 epochs. Under the same optimization settings, the test-set C-index increased further from 0.7770 to 0.8061, suggesting that the proposed temporal attention mechanisms continue to benefit from longer optimization schedules.

Due to computational limitations, equivalent 50-epoch experiments were not performed for all model variants, and therefore these results are not included in the primary comparison table. Nevertheless, this finding indicates that the reported results may underestimate the full performance potential of the proposed architecture under larger compute budgets.

5. Conclusion

In this work, we addressed the problem of predicting MCI-to-AD conversion from irregularly sampled longitudinal structural MRI data. While most prior deep learning approaches frame Alzheimer’s prediction as a static classification problem, our work focused on modelling temporal disease progression directly through the longitudinal MRI data.

We first established a longitudinal survival modelling baseline using a 3D CNN-transformer architecture trained under Cox partial likelihood. We then introduced two time-aware

improvements, each building upon the previous: continuous temporal encoding based on real elapsed visit times, and an ALiBi-inspired temporal attention bias that directly biases self-attention using inter-visit time gaps. Across the primary survival metrics, each temporal modification produced consistent gains. The final model achieved a C-index of 0.777, compared to the baseline’s 0.752, while substantially improving short-horizon prediction performance with an AUC@2y of 0.958.

Interestingly, binary classification metrics such as accuracy and F1-score remained unchanged across model variants. This suggests that the primary benefit of temporal modelling lies in improved risk ranking and temporal calibration rather than threshold-based classification. These findings support the hypothesis that explicitly modelling irregular clinical follow-up intervals provides clinically meaningful signal for disease progression prediction.

Despite these improvements, several limitations remain. The study was conducted on a relatively small cohort of 411 subjects, which may limit generalizability. Additionally, the improvements were evaluated cumulatively rather than through exhaustive ablation studies isolating each architectural component independently. Furthermore, results are reported from single training runs and may be sensitive to initialization and data partitioning. Our work also does not incorporate interpretability or explainability metrics, which remain important open problems in clinical deep learning systems.

Future work will focus on scaling the approach to longer longitudinal cohorts, incorporating multi-modal inputs such as PET and clinical biomarkers, and performing detailed ablation analyses of temporal attention mechanisms. Integrating techniques such as Grad-CAM or SHAP may further improve the clinical applicability and trustworthiness of longitudinal Transformer-based models for Alzheimer’s disease progression prediction. Additionally, due to computational limitations, all primary experiments were restricted to 30 training epochs; preliminary extended training experiments suggest that further performance gains may be achievable under larger compute budgets.

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